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Gaming disorder and its association with depression, social anxiety, and risk perception during the COVID-19 pandemic: A study using a Gaussian graphical model and moderated network models

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ABSTRACT

During the COVID-19 pandemic, many scholars in the field of behavioral addiction examined the risk of gaming disorder (GD). The association between GD, depression, social anxiety, and risk perception toward COVID-19 among Chinese university students has remained largely uninvestigated, especially using network analysis. Therefore, the present study (N = 1794) examined the relationship between these variables during the pandemic using Gaussian graphical model (GGM) and Moderated Network Model (MNM) approaches. In the GGM and MNM, GD had a significant interaction with depression. Individual risk perception and public risk perception had stronger connections in the network, as did depression and social anxiety. In addition, 'fatigue' was identified as the core symptom of depression. Neither moderation effects (i.e., three-way interaction between GD, depression, social anxiety, and risk perception) nor gender differences in network comparisons were found. These results suggest that relieving negative emotional states may have helped prevent GD during the COVID-19 pandemic, while the influence of risk perception on GD and negative emotions needs to be further examined.

1. Introduction

Many scholars in the field of behavioral addiction conducted research examining the risk of gaming disorder (GD) during the COVID-19 pandemic [1,2]. Gaming activities may reduce anxiety and stress, and provide opportunities to socialize [3,4], but a minority of individuals were found to be at greater risk of GD during the pandemic [1,2]. In the present study, the working definition of 'gaming' refers to videogames played on computers, consoles, and digital devices (e.g., smartphones, tablets) irrespective of whether they are played online or offline, and excluding both board games and gambling games, which is consistent with the World Health Organization's (WHO) description in relation to gaming disorder (i.e., 'digital gaming' or 'video-gaming' online/offline) [5,6].

As a persistent or recurrent maladaptive behavior, GD may influence mental and physical health (e.g., irritability and impulsiveness after have to stop playing videogames, and bad vision and cervical spine problems) [7], and even cause negative consequences (e.g., poor performance and interpersonal relationship and family conflicts) among a minority of individuals, especially adolescents and emerging adults [5,7]. The Disease Prevention and Control Bureau of China (DPCBC) reported the prevalence of GD to be approximately 5 % [7], which is higher than the 2–3 % reported pooled prevalence around the world [8]. The different prevalence of GD may be associated with country/cultural context [8]. One systematic review reported a significant increase of GD among children and adolescents during the pandemic [2]. In addition, the importance of monitoring and prevention of GD during the pandemic and its later stages was proposed by some researchers [9]. There are still many debates regarding the influencing factors of GD and interactional effect of multiple factors, especially during the pandemic [10,11].

The Interaction of Person-Affect-Cognition-Execution (I-PACE) model posits that multiple variables including an individual's core characteristics, affective mood states, cognition, execution, and environmental factors (e.g., social distancing and home quarantine during the COVID-19 pandemic) may interact to lead to addictive behaviors (e.

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g., gaming disorder) [12,13]. The results of Teng et al.'s study supported I-PACE theory and found that individuals' responses to COVID-19 (i.e., risk perception) mediated the association with psychological affect (i.e., anxiety and depression) and internet gaming disorder (IGD) [14]. According to the I-PACE model, some risk factors (e.g., negative emotions, distorted cognitions, and inappropriate coping styles) may lead to GD [15,16]. In addition, perceived risk of COVID-19 may cause negative mood states (e.g., anxiety and depression) [17], and negative emotions were found to influence IGD, which is consistent with the I-PACE model [18]. GD may also stabilize and intensify negative emotions and cognitive components in the process of maintaining addictive behaviors based on the I-PACE model [12].

Superior to traditional statistical methods such as correlation analyses and *t*-tests, network analysis is able to methodologically study the organization of the whole system's components and visually display (in color) the relationship between multiple variables in lots of different science fields [19]. Nodes (i.e., study variables) and edges (i.e., connections between two variables) are identified as the basic concepts in network analysis [20]. Thicker green edges signify more significant positive associations [20]. In network models, the Gaussian graphical model (GGM) has been used widely in social science fields in recent years [21,22], while moderated network models (MNMs) have only been used since 2021 [23].

The Gaussian graphical model (GGM) is considered as an undirected and powerful probabilistic network of partial correlation coefficients [22], which has been used extensively for exploring the core symptoms of various mental health disorders and potential causal relationships [22,24–26]. The moderated network model (MNM) refers to "each pairwise interaction between two variables to be moderated by (a subset of) all other variables in the GGM [23]. The GGM assumes each pairwise interaction is independent when other variables are conditional [22]. Such moderating or mediating effects often occur in psychological and social research [23]. The MNM is also suitable for comparing network differences for more than two groups [27]. Tao et al. used the MNM and found that attention to negative information (ANI) was the moderator between smartphone addiction, fear of missing out (FoMO), and depression [28]. Therefore, the GGM and MNM may be utilized to better explain the interactional and complex association between GD, negative emotions, and risk perception during the COVID-19 pandemic.

2. Related research and study hypotheses

A systematic literature review of studies published from 2010 to 2020 concluded that there was a strong association between IGD and social anxiety [29]. Problematic gaming was also associated with social anxiety during the COVID-19 pandemic [30]. Another meta-analysis of studies carried out during the pandemic reported that gaming may result in short-term relaxing effects and stress relief for most individuals, but cause detrimental effects, such as higher anxiety, depression, and GD for a minority of individuals [31]. Some scholars have also reported that GD (in both online and offline forms, and in both single player and multiplayer formats) is associated with social anxiety and loneliness based on motivations of need satisfaction [32,33]. Moreover, anxiety and depression have been found to mediate the association between GD and insomnia during the pandemic [34]. Based on the I-PACE model and the aforementioned studies, it was hypothesized that there would be a significant interaction between GD, depression, and social anxiety among Chinese university students during the COVID-19 pandemic using the GGM approaches (H_1).

Risk perception refers to individual or public perceived susceptibility to threats, such as diseases and natural disasters [35,36]. Cori et al. indicated that during the COVID-19 pandemic, risk perception may have influenced individuals' beliefs, attitudes, knowledge, decisions, and behaviors [37]. One study investigating an epidemiological model reported a reduction in risk perception when an infection wave reoccurred [38]. In addition, increasing screen time (e.g., gaming and social media use), high acceptance of health-related behaviors to inhibit the spread of COVID-19 (e.g., washing hands, physical distancing, etc.), relatively low risk perception, and low fear of infection were reported during the pandemic [39]. Based on the I-PACE model and the aforementioned studies, it was hypothesized that there would be a significant interaction between GD and risk perception (i.e., a cognitive factor) among Chinese university students during the COVID-19 pandemic using the GGM approaches (H_2).

In a previous study, individual risk perception during the COVID-19 pandemic was associated with subjective judgment and intuitive experience, which also influence public risk perception and may cause social anxiety and public panic [40]. Individual subjective judgments to risky events (i.e., risk perception) are inevitably associated with feelings (e.g., anger, anxiety, disgust, etc.) [41]. In one study, the association between risk perception and mental health during the COVID-19 pandemic was mediated by negative emotions [42]. In addition, the perceived vulnerability to COVID-19 (i.e., risk perception) mediated the relationship between exposure to COVID-19 news and depression, which indicated a positive association between greater risk perception and depression [43].

In a South African survey study, the increase of risk perception toward COVID-19 appeared to predict greater depression among adults after four months of the COVID-19 breakout [44]. Individuals with greater risk perception toward COVID-19 also reported higher anxiety [45,46]. Zhao et al. also reported that risk perception significantly associated with anxiety and mediated the relationship between positive information of COVID-19 and anxiety among Chinese students [47]. Based on the I-PACE model and the aforementioned studies, it was hypothesized that there would be a significant interaction between risk perception, depression, and social anxiety among Chinese university students during the COVID-19 pandemic using GGM approaches (H_3).

Tripp et al. noted that highly anxious individuals may perceive negative events more severely, which indicates a close association between risk perception and anxiety [48]. Risk perception has also been reported as a mediator between the period of attention to COVID-19 news and depression [49]. Another study reported that knowledge regarding COVID-19, anxiety, depression, and risk perception toward COVID-19 may influence preventive behavior [50]. Moreover, high risk perception of COVID-19 has been found to moderate the relationship between positive emotions and psychological well-being among gamers (i.e., gamers with higher risk perception of COVID-19 had lower positive emotions and poorer psychological well-being during the COVID-19) [51]. In addition, negative metacognitions (e.g., "uncontrollability and dangers of online gaming") have been found to mediate the relationship between social anxiety and IGD [52]. Based on the aforementioned studies, it was hypothesized that risk perception (i.e., a cognitive factor) would moderate the relationship between depression and anxiety (i.e., affective factors) and GD among Chinese university students during the COVID-19 pandemic using MNM approaches (H_4).

A previous study reported that male gender, anxiety, loneliness, and high mentalization predicted disordered gaming during the COVID-19 pandemic [53]. Moreover, social anxiety was found to mediate the relationship between relational victimization and gaming addiction among female university students during the COVID-19 pandemic [54]. Based on males being at greater risk of GD than females [55,56], it was hypothesized that there would be significant gender differences in the network structure and global strength using a GGM (H_5).

3. Materials and methods

3.1. Participants and procedure

Using convenience sampling, a cross-sectional online survey was conducted via the *Wenjuanxing* online platform among 1883 Chinese university students from five provinces (i.e., Jiangxi [n = 892], Liaoning [n = 313], Shannxi [n = 265], Heilongjiang [n = 348] and Xinjiang [n =

65]) after the COVID-19 break out in China at the beginning of 2020. A QR code that linked to an online survey was sent to teachers who were responsible for students' management or teaching at different universities (using the research team's personal contacts and networks). The teachers then sent the QR code to their students inviting them to participate in a survey. Data collection took place during the whole of the winter holiday from 28th January to 10th March (school holidays of provinces were different due to climatic reasons and the impact of the epidemic). The online survey with the purpose of the study was sent to participants and their informed consent was obtained. The inclusion criteria for participants were (i) being a Chinese university student, (ii) having played videogames in the past 12 months, and (iii) volunteering to participate the study. The exclusion criteria were (i) being a non-gamer (no playing videogames in the past 12 months) and (ii) being aged under 18 years. All items needed to be completed for the online data to be submitted successfully. Therefore, there were no missing data. Due to short response times (i.e., less than 60 s), 89 participants were excluded. Of the remaining 1794 participants, 1339 videogame players (652 males and 687 females; age range 18 to 25 years old; mean age = 19.1 years [SD = 1.7]) were retained for analysis.

3.2. Measures

3.2.1. Demographic questionnaire

Demographic information was collected included gender, age, residential status (i.e., urban or rural), played any videogame in the past 12 months, weekday gaming hours, and weekend gaming hours.

3.2.2. Gaming Disorder Test (GDT)

The GDT was developed by Pontes et al. and items are based on the criteria for gaming disorder in ICD-11 [57]. The GDT includes four items which are rated on a five-point Likert scale from 1 (*Never*) to 5 (*Always*). In addition, the scale has been verified as a good assessment tool for GD among Chinese university students [57,58]. Potential or high-risk indication of GD was based on at least one of the scale items being answered 'often' or 'always' [57]. In the present study, the Cronbach's alpha and McDonald's ω of the GDT were 0.867 (CI: 0.855–0.878) and 0.874 (CI: 0.862–0.885), which indicated that the GDT had very good internal consistency.

3.2.3. Patient Health Questionnaire-9 (PHQ-9)

The nine-item Patient Health Questionnaire-9 (PHQ-9) was developed using the criteria of depressive disorders in the *Diagnostic and Statistical Manual of Mental Disorders* [59] and has been validated among Chinese university students [60]. Items are rated on a Likert scale from 0 (*Not at all*) to 4 (*Nearly every day*). A higher score indicates more severe depression. In the present study, the Cronbach's alpha and McDonald's ω of the PHQ-9 were 0.895 (CI: 0.886–0.903) and 0.899 (CI: 0.890–0.907), which indicated that the PHQ-9 had very good internal consistency.

3.2.4. Short version of Social Interaction Anxiety Scale (SIAS)

Fergus et al. developed the short version of Social Interaction Anxiety Scale (SIAS) [61]. The SIAS has been verified has having good validity and reliability among Chinese university students [62]. The SIAS includes six items which are rated on a five-point Likert scale from (*"Not at all a characteristic or true of me"*) to 5 (*"Extremely characteristic or true of me"*). Higher scores indicate a greater level of social anxiety. In the present study, the Cronbach's alpha and McDonald's ω of the SIAS were 0.820 (CI: 0.805–0.834) and 0.853 (CI: 0.840–0.865), which indicated that the SIAS had very good internal consistency.

3.2.5. The Risk Perception Questionnaire (RPQ)

The Risk Perception Questionnaire (RPQ) was developed by Yan and Wen [63]. The RPQ has eight items comprising two dimensions (each with four items): individual risk perception (e.g., *"The COVID-19 outbreak is closely related to me"*) and public risk perception (e.g., *"The COVID-19 outbreak affects the whole country"*). Participants rate each item from 1 (*Totally disagree*) to 5 (*Totally agree*). The Cronbach's alpha and McDonald's ω of the total scale were 0.772 (CI: 0.753–0.789) and 0.772 (CI: 0.753–0.790), 0.622 (CI: 0.590–0.652); 0.663 (CI: 0.634–0.693) for the individual risk perception subscale; and 0.744 (CI: 0.722–0.765) and 0.767 (CI: 0.747–0.787) for the public risk perception subscale, which indicated that the RPQ had good internal consistency.

3.3. Data analysis

Descriptive statistics (i.e., means, standard deviations, frequencies, and percentages) and group comparisons (i.e., *t*-test and χ^2 test) were conducted using SPSS 20.0, while the reliability (Cronbach's alpha and McDonald's $\omega \geq 0.7$ being the accepted threshold), and GGM and MNM were performed using R 4.3.2. The network comparison test (NCT) was also conducted using the "NetworkComparisonTest" package.

The GGM was performed based on the Extended Bayesian Information Criterion LASSO (EBICglasso) [26], which represented network characteristics through nodes and edges. The tuning parameter was set to 0.5 considering sensitivity and specificity. The network accuracy and stability were tested through the edge-weight accuracy and testing for significant differences of nodes and edges (i.e., a non-parametric bootstrap with 1000 samples), and centrality stability (i.e., a case-dropping subset bootstrap), while the correlation stability coefficient (CS-coefficient, ≥ 0.25) was calculated to test the node centrality stability [22]. The MNM was estimated according to a l_1-regularized nodewise regression method [23]. The R-package *mgm* was used to calculate the MNM. In addition, all variables (i.e., GD, depression, anxiety, individual risk perception and public risk perception) were set as moderators ("moderators = 1:5") and mainly explored the moderating effect of risk perception between GD and depression/anxiety. All predictors were scaled to mean zero and standard deviation to avoid parameter penalization [23]. Finally, the network comparison test (NCT) was performed to calculate the network structures and global strengths across gender.

3.4. Ethics

The Gannan Medical University Research Ethics Committee approved the study (TD201906) in accordance with the Declaration of Helsinki and informed consent was obtained from all participants.

4. Results

4.1. Descriptive/correlation analysis

There were 115 participants (8.6 %) classed as being at high-risk of GD (i.e., at least one of the four scale items being answered 'often' or 'always') comprising 70 males (10.7 % of 652 males) and 45 females (6.6 % of 687 females) which was significantly different ($\chi^2 = 7.466$; $p = 0.006$) across gender. Males had significantly higher total GDT score (M = 7.41) and longer gaming time than females (M = 6.03, all *p*-values < 0.001 and strong effect sizes [i.e., Cohen's *d*/Phi], see Table 1). There were no significant differences on any variables regarding residential status. In addition, GD had statistically significant correlations with depression and social anxiety (Table 2).

4.2. Gaussian graphical model (GGM)

The GGM networks including GD, depression, social anxiety, and risk perception are shown in Fig. 1 (A-C: factor-level network and D-F: item-level network). In the factor-level GGM network, stronger edges were identified including the public risk perception and individual risk perception ($r = 0.551$ for 1339 participants, $r = 0.519$ for 652 males, $r = 0.474$ for 687 females), and depression and social anxiety ($r = 0.456$ for 1339 participants, $r = 0.343$ for 652 males, $r = 0.454$ for 687 females),

Table 1
Descriptive characteristics of the sample (n = 1,339).

Variables	Total (n = 1339)	Gender		t/χ^2	p	Cohen's d/ Phi	Residential status		t/χ^2	p	Cohen's d/ Phi
		Males (n = 652)	Females (n = 687)				Urban (n = 620)	Rural (n = 719)			
Age (years)	19.0 ± 1.7	19.0 ± 1.7	18.9 ± 1.7	0.66	0.508	0.036	19.0 ± 1.7	19.0 ± 1.7	1.60	0.109	0.089
Gaming disorders	6.70 ± 2.77	7.41 ± 2.83	6.03 ± 2.53	9.37	<0.001	0.512	6.53 ± 2.69	6.86 ± 2.82	2.19	0.029	0.120
Depression	13.09 ± 4.17	12.89 ± 4.12	13.28 ± 4.20	1.71	0.088	0.093	13.17 ± 4.24	13.02 ± 4.10	0.65	0.513	0.036
Social anxiety	15.94 ± 5.31	15.66 ± 5.23	16.20 ± 5.38	1.86	0.063	0.102	15.83 ± 5.29	16.03 ± 5.33	0.70	0.485	0.021
Risk perception (RP)	31.55 ± 4.77	31.48 ± 5.02	31.61 ± 4.52	0.52	0.603	0.028	31.64 ± 4.80	31.46 ± 4.75	0.68	0.498	0.037
Individual's RP	14.63 ± 3.07	14.51 ± 3.26	14.73 ± 2.88	1.31	0.191	0.072	14.66 ± 3.11	14.60 ± 3.03	0.38	0.701	0.021
public RP	16.92 ± 2.31	16.96 ± 2.37	16.88 ± 2.25	0.67	0.506	0.036	16.98 ± 2.32	16.87 ± 2.30	0.89	0.373	0.049
Weekday gaming hours											
≤3	1194 (89.2 %)	556 (86.1 %)	638 (95.2 %)	20.32	<0.001	0.123*	555 (89.5 %)	639 (88.9 %)	0.14	0.931	0.010
3 ~ 10	125 (9.3 %)	84 (12.2&)	41 (4.0 %)				56 (9.0 %)	69 (9.6 %)			
≥10	20 (1.5 %)	12 (1.7 %)	8 (0.8 %)				9 (1.5 %)	11 (1.5 %)			
Weekend gaming hours											
≤3	1019 (76.1 %)	430 (66.0 %)	589 (85.7 %)	72.53	<0.001	0.233*	471 (76.0 %)	548 (76.2 %)	0.46	0.793	0.019
3 ~ 10	302 (22.6 %)	208 (31.9 %)	94 (13.7 %)				142 (22.9 %)	160 (22.3 %)			
≥10	18 (1.3 %)	14 (2.1 %)	4 (0.6 %)				7 (1.1 %)	11 (1.5 %)			

Table 2
Correlation analysis of the study variables.

Variable	GD	Dep	SA	**RP**	IRP	PRP
GD	1.00					
Dep	0.361**	1.00				
Log (BF_{10})	89.841					
SA	0.244**	0.511**	1.00			
Log (BF_{10})	37.740	198.200				
RP	−0.003	0.057*	0.079**	1.00		
Log (BF_{10})	−3.368	−1.223	0.857			
IRP	0.018	0.072**	0.089**	0.916**	1.00	
Log (BF_{10})	−3.147	0.065	1.880	∞		
PRP	−0.031	0.022	0.046	0.847**	0.563**	1.00
Log (BF_{10})	−2.748	−3.056	−1.934	∞	250.784	

Note: GD = gaming disorder; Dep = depression; SA = social anxiety; RP = risk perception; IRP = individual risk perception; PRP = the public risk perception.

and GD and depression ($r = 0.277$ for 1339 participants, $r = 0.309$ for 652 males, $r = 0.218$ for 687 females). The depression node had the strongest centrality (strength = 1.289 for 1339 participants, strength = 1.705 for 652 males, strength = 1.229 for 687 females) (Appendices S1–S5).

The edge-weight accuracy should be explained carefully owing to the part wide bootstrapped CIs, while the CS-coefficient (CS [cor = 0.7] = 0.75) indicated better centrality stability (Figs. 2A and B) in the factor-level network. The strongest edge 'irp' (i.e., individual risk perception) and 'prp' (i.e., public risk perception) was significantly different from other pairwise of edges. Five nodes' strengths were also significantly different from one another (Figs. 2C and D). In the item-level GGM network, the item dep 4 (*"Feeling tired or having little energy"*) had the strongest centrality (1.429) (Appendix S6).

4.3. Moderated network model (MNM)

The values of pairwise interaction were 0.51 between the individual risk perception and public risk perception, 0.44 between depression and social anxiety, 0.26 between depression and GD, and 0.04 between GD and social anxiety, while risk perception (i.e., individual risk perception and public risk perception) had no interaction with GD, depression, and social anxiety (Fig. 3). In addition, no moderation effect (i.e., three-way interaction) was found in the present study.

4.4. Network comparison between gender

The factor-level ($M = 0.167$, $p = 0.201$) and item-level ($M = 0.135$, $p = 0.537$) network structures had no significant gender differences via the NCT. Additionally, the factor-level (1.35 vs. 1.46, $p = 0.509$) and item-level (11.4 vs. 11.1, $p = 0.257$) global strengths had no significant gender differences.

5. Discussion

The present study examined the network structure of GD, depression, anxiety and risk perception during COVID-19 among a relatively large sample of Chinese university students, using the moderating network approach. In the GGM and MNM, GD had a significant interaction with depression, which supported the I-PACE model. Individual risk perception and public risk perception had stronger connections in the network, as did depression and social anxiety. In addition, 'fatigue' was identified as the core symptom of depression. These results suggest that interventions are needed to prevent or treat negative mood states.

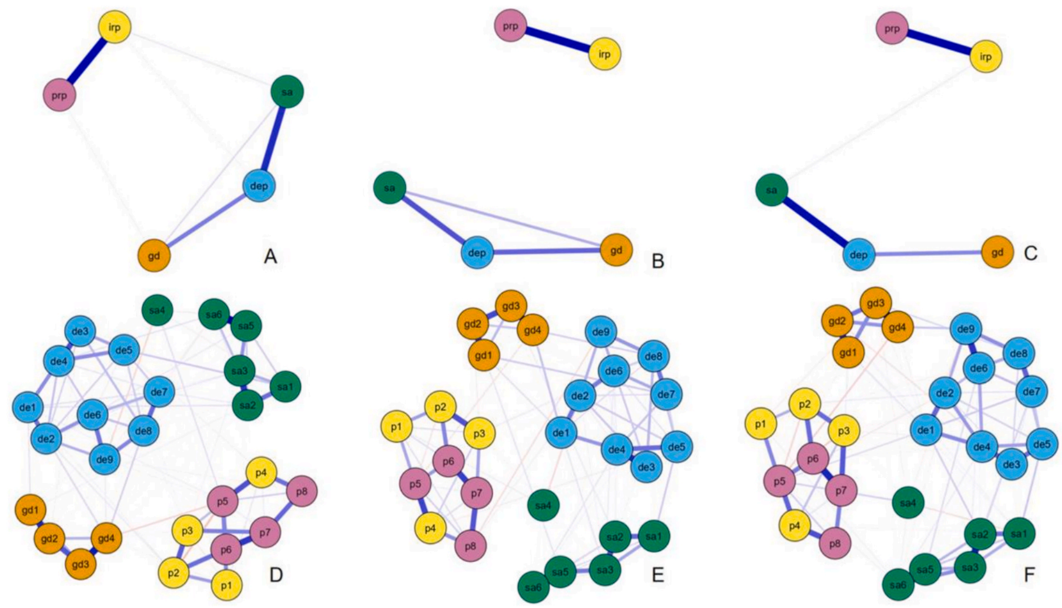

Fig. 1. EBICglasso model based on network analysis according to the relationship between variables among total participants (A) & (D), males (B) & (E), and females (C) & (F). Note: gd, gd1 ~ gd4 = gaming disorder; dep, de1 ~ de9 = depression; sa, sa1 ~ sa6 = social anxiety; irp, p1 ~ p4 = individual's risk perception; prp, p5 ~ p8 = the public risk perception.

Although these relationships were identified during the pandemic, they are likely to occur with GD more generally, irrespective of the context. Relieving negative mood states may help prevent GD, while the influence of risk perception on GD and negative emotions needs to be further examined.

In the present study, the prevalence of those at risk of GD (based on the Pontes et al.'s criteria [57]) was 8.6 %. This suggests that university students were at high risk of GD during the COVID-19 pandemic, which was consistent with the previous study where the prevalence of probable IGD was higher at the beginning of COVID-19 pandemic than before the COVID-19 pandemic [64]. However, this did not represent the true prevalence of GD but was simply an estimate of those at high-risk of GD. In another study, Pontes et al. used more stringent criteria to estimate the prevalence of GD [65]. In that study, all four scale items needed to be answered 'often' or 'always' (rather than just a single item). Using this more stringent method results in much lower prevalence estimates of GD (1.96 % in the study by Pontes et al. using samples from gaming communities) [65].

The DPCBC reported that adolescents are vulnerable to gaming disorder which can lead to many negative outcomes such as refusing to go to school or dropping out of school due to gaming [7]. A recent meta-analysis also reported that children and adolescents had the highest prevalence of GD (6.6 %) compared to adults (1.9 %) [8]. Moreover, significant correlations in the present study between GD, depression, and social anxiety were found, which is consistent with previous research [14,34,66]. The correlations between risk perception and GD, depression, and social anxiety were not supported in the Bayesian factor analysis. Therefore, the relationships between these variables needs further exploration.

In the GGM, individual risk perception and public risk perception, depression and social anxiety, and GD and depression had stronger edges, which indicated good structure validity of risk perception and a closer relationship between negative emotions (i.e., depression and social anxiety), and supported the I-PACE model. Wang et al. also indicated that public risk perception was influenced by factors including individual risk perception (i.e., understanding and perception of the COVID-19 pandemic) during the COVID-19 pandemic [67]. Due to being a collectivist culture, the Chinese public's risk perception also influences individual risk perception (i.e., the public believe in the government's ability to cope with emergencies and trust the power of medical technology, which convey a sense of security to individuals). Therefore, the edge of individual risk perception and public risk perception showed a strong association among Chinese university students.

In clinical practice, depressive disorder and anxiety disorder often occur comorbidly [68,69], while depression and anxiety as negative emotions may also arise with other psychosomatic disorders [70,71]. During the COVID-19 pandemic, symptoms of depression and anxiety significantly increased with other physical problems [72,73]. In the present study, GD also had a significant interaction with depression, which was consistent with previous research [74,75].

Yang et al. reported a close bridge connection between internet gaming disorder (IGD) and anxiety among a sample of Chinese children during the COVID-19 lockdown, and reported social phobia as the central node in the IGD and anxiety network [76]. However, a statistically significant reciprocal association between GD and social anxiety was not found in the present study (Fig. 1, A-C), which may be due to different participants (i.e., Yang et al.'s participants were from the first year of primary school in China) and measurement tools (i.e., Yang et al. used the Spence Children's Anxiety Scale [77] and the Internet Gaming Disorder Scale-Short Form [78]). Therefore, the results partly supported H_1, because there was an interaction between GD and depression, but not GD and social anxiety.

Engaging in gaming for social needs during the pandemic may have decreased negative emotional experience (e.g., self-isolation and social anxiety) [79]. When individuals are immersed in gaming, especially massively multiplayer online role-playing game [MMORPGs] and real-time strategy games, the risk perception toward COVID-19 may be neglected or forgotten temporarily. One of the characteristics of GD is *"increasing priority given to gaming"* [5], which also means a stronger focus on gaming rather than the gamer's surroundings and other daily activities. In previous research, the associations between GD/IGD and risk-taking, risk evaluation, or risk making-decision have been examined [80–82], but the relationship between risk perception and GD in these studies remained unclear.

The I-PACE model refers to the cognitive component as one of the important factors in the development and maintenance of GD (i.e., early and late stages). However, cognition toward the unprecedented context of COVID-19 (i.e., risk perception) was not as important as either the

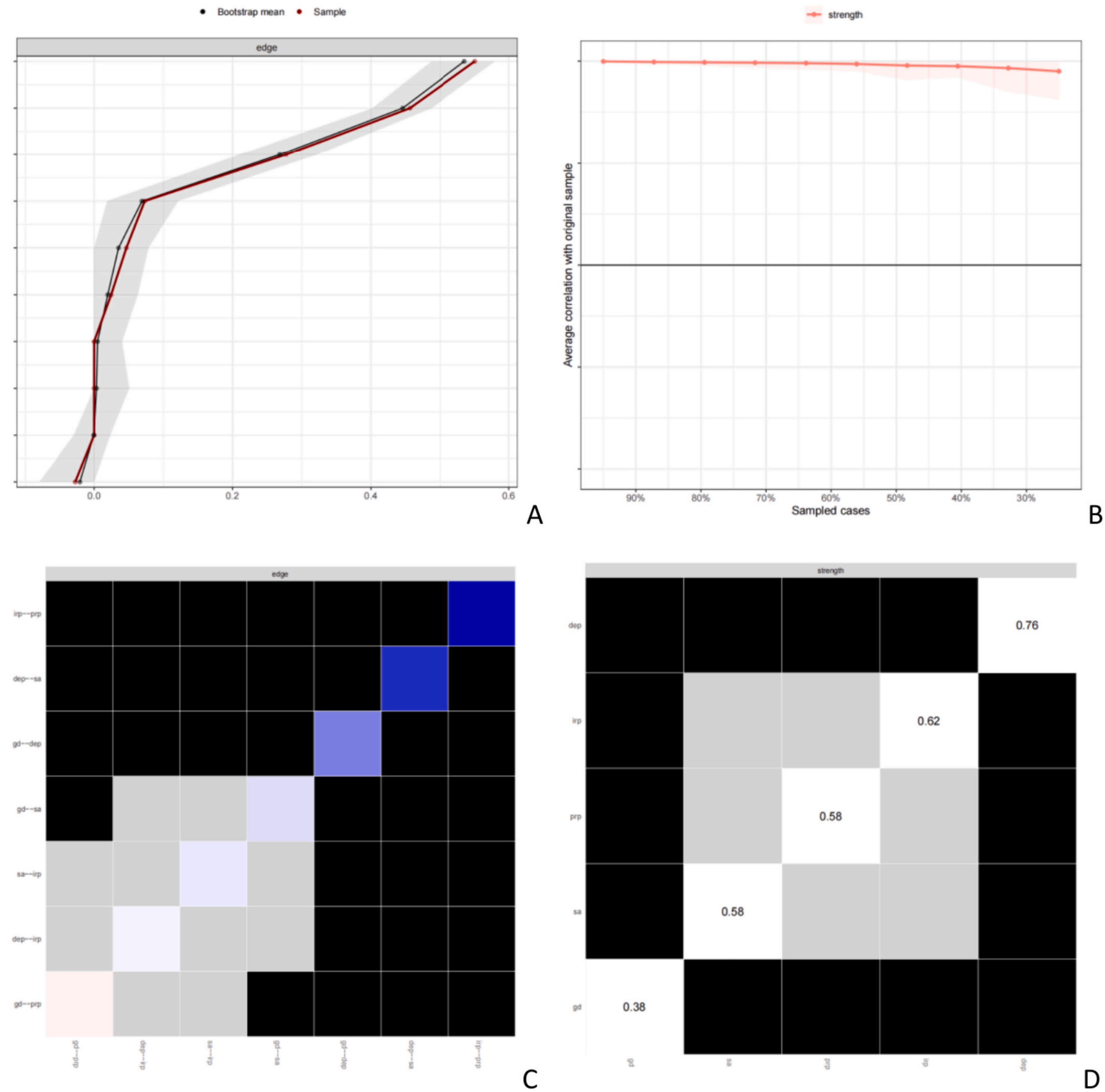

Fig. 2. Bootstrapped confidence intervals of estimated edge-weights (A) and Case-dropping bootstrap procedure for node strength (B), Bootstrapped difference tests ($\alpha = 0.05$) between edge-weights that were non-zero in the estimated network (C) and node strength of the items (D).

cognition of gaming itself or gaming motivation in the present study. This may be because a minority of participants at high risk of GD paid more attention to playing videogames for relaxation and relieving negative emotion rather than risk perception towards COVID-19 during the pandemic. Therefore, H_2 was not supported because the reciprocal association between risk perception and GD was not significant in the present study.

The depression node was found to have the strongest centrality, especially the item *"Feeling tired or having little energy"* (i.e., 'fatigue') which was identified as the core symptom of depression. This is the same node as other studies examining Chinese populations [83–85]. The accuracy and stability of the GGM were examined through edge-weights, node centrality, and testing for significant differences of nodes and edges. This verified the structure and characteristics of the network including GD, depression, social anxiety, and risk perception.

Olagoke et al. reported that the perceived vulnerability to COVID-19 (i.e., risk perception) mediated the relationship between exposure to COVID-19 news and depression, which indicated a positive association between greater risk perception and depression [43]. However, the significant reciprocal effects between depression and risk perception, and between social anxiety and risk perception were not found in the present study. Therefore, H_3 was not supported, perhaps due to the different sample and the different measurement tools (i.e., Olagoke et al.'s participants were resident in the United States, reported a mean age of 32.44 years [SD $\pm$ 11.94], and utilized the 2-item Patient Health Questionnaire [PHQ-2] and the 3-item Perceived Vulnerability to

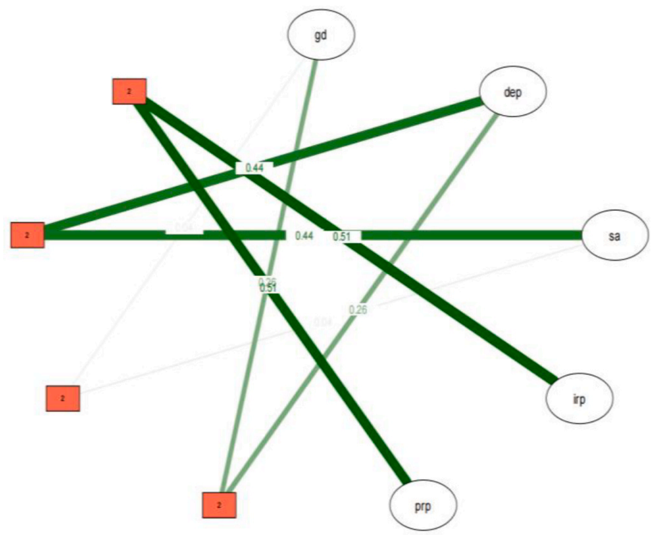

Fig. 3. MNM network analysis among total participants. Red square nodes represented pairwise interaction. Note: gd = gaming disorder; dep = depression; sa = social anxiety; irp = individual's risk perception; prp = the public risk perception. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

COVID-19 Scale [43]).

In a longitudinal study, Teng et al. reported that risk perception toward COVID-19 mediated the relationship between depression/anxiety and IGD [14]. The present study's survey was completed during the winter holiday, therefore, the students who lived with their family may have (i) felt safer and more relaxed, (ii) communicated freely online with their friends, classmates or relatives via the social media platforms (e.g., *WeChat* and *Tencent QQ*), and (iii) been subject to home quarantine. All of these factors may have decreased social anxiety and depressive symptoms. Therefore, risk perception towards COVID-19 may weaken the association between the variables (i.e., risk perception, anxiety, and depression). There was no significant interaction between risk perception, anxiety, and depression using the MNM in the cross-sectional study. Therefore, H_4 was not supported. Moreover, there were no gender differences in the network structure or global strength. Therefore, H_5 was not supported.

There are several limitations to be noted. First, a self-report survey may be influenced by participants' memory recall and social desirability. Second, the convenience sampling and personal networks may not have been representative of the Chinese university student population. Third, cross-sectional data are unable to help in the determination of causality and temporal changes between the study variables (i.e., GD, depression, anxiety and risk perception). Fourth, risk perception toward COVID-19 as single cognitive component may have weaker influence on GD than other cognitive components (e.g., ruminative thoughts about gaming and harmful beliefs about gaming). Fifth, game genre (e.g., MMORPGs and real-time strategy games) was not be examined as single variable in the present study. Sixth, the increase in GD during the pandemic, especially due to lockdowns in all countries, may have resulted in an increase of false-positives in relation to gaming disorder. Moreover, the restrictions in China were typically stricter than in Western countries because of the large (1.4 billion) population size and challenges related to an unbalanced distribution of healthcare resources. In the collectivist cultures, the public's cooperation with epidemic and pandemic prevention is likely to be higher than individualistic cultures, which reduces the resistance to policy implementation.

6. Conclusions and future directions

In summary, the present study examined the reciprocal relationships between depression and anxiety, and between depression and GD among a sample of Chinese university students during the COVID-19 pandemic. The significant interaction of GD and depression was verified using GGM and MNM approaches. In addition, depression was found to have the strongest centrality in the network, while the item *"Feeling tired or having little energy"* (i.e., 'fatigue') was identified as the core symptom of depression.

These findings have broader social and policy implications. Gamers should receive educational awareness regarding the potentially addictive nature of GD, and have access to professional psychological counseling and treatment should they need it. Research teams in the area based in universities should provide referrals for their students to practitioners so that those experiencing problems can receive the help and support they need for GD, such as cognitive-behavioral therapy and/or group counseling to relieve negative emotions associated with GD [86–89]. Governments should also should regulate the video gaming industry to help in the development of safer gaming tools (similar to that of gambling operators) such as limit-setting, voluntary self-exclusion, mandatory play breaks, and personalized in-game messaging. The study's findings offer insights beyond a pandemic-specific context because the negative and aversive mood states associated with GD occur in non-pandemic contexts.

Future research should use more objective methods (e.g., neuroscience approaches and clinical diagnoses) and more representative sampling (e.g., random sampling, recruitment of participants from online gaming forums or communities). Longitudinal studies are needed to better understand the interaction between the study variables examined here. Future studies could also include diverse age groups or populations (e.g., from children through to elderly populations, or more multicultural populations from different countries) to enhance the generalizability and provide comparative insights into how GD affects mental health across different demographics. Moreover, the cognitive components closely associated with GD should also be further examined, such as dysfunctional or distorted beliefs toward gaming behaviors. In addition, the influence of risk perception on GD and negative emotions needs to be further examined.

Consent for publication

Not applicable.

CRediT authorship contribution statement

Li Li: Writing – original draft, Formal analysis, Conceptualization. **Zhimin Niu:** Writing – original draft, Investigation, Conceptualization. **Xi Gong:** Investigation. **Zhiyu Pi:** Investigation. **Songli Mei:** Methodology. **Mark D. Griffiths:** Writing – review & editing.

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Declaration of competing interest

There are no financial or non-financial competing interests except for MDG. MDG has received research funding from *Norsk Tipping* (the gambling operator owned by the Norwegian government). MDG has received funding for a number of research projects in the area of gambling education for young people, social responsibility in gambling and gambling treatment from *Gamble Aware* (formerly the *Responsibility in Gambling Trust*), a charitable body which funds its research program based on donations from the gambling industry. MDG undertakes consultancy for various gambling companies in the area of player protection and social responsibility in gambling. None of the research staff received incentives for recruiting participants or for any other purpose directly associated with the study.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.entcom.2025.101024.

Data availability

Data will be made available upon reasonable request to the corresponding author.

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